

Skills and Resources Audit

Resources:

Clean Ubuntu 22 install:

A bare version of Ubuntu 22 installation with all the dependencies installed. Used for final testing and deployment, separate from the development VMs. This matters because ARM64 is the amd64 architecture causes discrepancies and the code needs validating on real, clean environments before ensuring success.

Ubuntu 22 VMs:

Virtual machines running Ubuntu 22 that most team members can run day to day to develop. These VMs are needed so the team can work consistently in the same Linux environment regardless of Host OS. Avoids the whole “works on my machine” issue.

Unitree SDK (Python) Repository:

This is the official Software Development Kit to communicate with the G1 robot. Necessary as the core interface between our code and the robot’s physical response. Will be used for both simulated (using Mujoco) and real control.

Unitree Mujoco Repository:

The official simulation program for testing code for the G1 before real-world testing. Let’s the team test motion, kinematics, reaction logic etc. before risking it on the real G1 robot. All gesture response and mirroring reactions validated here before real-world deployment.

MediaPipe Repository:

Pose/gesture detection library developed by Google. Will be used to process camera video into skeleton and gesture information. Used to identify the leader and reading their gestures. It’s output feeds the decision making process.

Pinocchio Repository:

A “rigid-body dynamics library” to compute the G1 robot’s kinematics (joints, poses, movements). Translates desired action into valid commands for the robot’s joints.

GMR Repository:

Tool for converting motion-capture suit data into robot motion data.

Github:

GUI for Git repository to host team’s project on. Will include Setup and Use README. Also hosts all team documentation.

Microsoft Teams:

Teams primary source of communication with client. Used to deliver weekly deliverables for project and Minutes to the client.

WhatsApp:

The secondary lightweight communication channel for quick messages. Used to avoid cluttering the formal Teams conversation with quick, informal messages.

Unitree G1:

The physical humanoid robot the project is based around. The end target for all software. Software will drive this hardware.

Mo-Cap Suit:

The physical motion capture suit used to generate supplementary motion data. Needed to test the GMR skeleton pipeline and validate robot's mirroring ability to the motion capture suit

VS Code:

Agreed IDE. Used for consistent development experience. Easier to debug when everyone is using the same tool.

*Skills:***Python Programming:**

The primary language for the SDK, for MediaPipe and Pinocchio which also uses Python calls. Additionally will control most of the decision and control logic. Essential for every deliverable in the project.

Documentation Comprehension:

The ability to read and comprehend repository documentation and then apply that to the project. The project is almost entirely built upon integrating external third party tools rather than writing everything from scratch. As such it is vital to be able to read and comprehend the documentation associated with the dependencies.

Robotics Theory:

Understanding joint movement, degrees of freedom and how commands translate to physical motions. A basic understanding is required to use and incorporate the Pinocchio repository as well as understand realistic robot behaviour.

Version Control / Git:

Practical skill in using the Git framework to maintain a good version control system. Team should be adept in branching off commits and merging back.

Computer Vision / Pose Estimation:

Understanding how MediaPipe extracts skeletons, bounding boxes and gestures from video input. Important for projects' "sensing" requirements.

Linux/Ubuntu System Administration:

Basic ability to install, configure and troubleshoot Ubuntu problems across VMs and clean installs. This can also mean navigating amd64 and ARM64 differences. Entire

team must independently get a working development environment before any code can be tested.

Agile / Project Management:

Familiarity with the Agile Development framework. Sprints, stories, roles based task division, etc.